

Complex Analysis PhD Exam, September 2009

1. A fixed point of a function f is a point x with $f(x) = x$. Let G be an open, simply connected subset of $\mathbb{C}$. If f is analytic on G , $f(G) \subset G$, and f has two fixed points, show that f is the identity map.
2. Let G be an open, connected subset of $\mathbb{C}$ and f be analytic on G . Note that we are not assuming that G is simply connected.
 - (a) Assume now that for another open, connected set H , $f(G) \subset H$ and $h : H \rightarrow \mathbb{R}$ is harmonic. Show that $h \circ f$ is harmonic.
 - (b) Prove or disprove: Both the real and imaginary parts of f are harmonic functions.
 - (c) Prove or disprove, if $u : G \rightarrow \mathbb{R}$ is harmonic, then there is a harmonic function $v : G \rightarrow \mathbb{R}$ so that $u + iv$ is analytic.
3. Give an explicit biholomorphism from the “wedge” $\{z : 0 < \arg(z) < \pi/2\}$ onto the open unit disk.
4. Evaluate the integral, and justify each step.

$$\int_{-\infty}^{\infty} \frac{x \sin x}{(x^2 + 1)^2} dx.$$

5. Let f be an entire function of finite order. Prove that if the order is not an integer, then f must have infinitely many zeros. Does there exist an entire function of infinite order with finitely many zeros? Explain.
6. **Warning: Suspected error.** The statement is preserved as printed, but it overlooks the possible case $g \equiv 0$, for example when $g_n \equiv 1/n$. The intended correction is not uniquely determined: one could assume $g \not\equiv 0$ or conclude that g is identically zero or has only real zeros.

Assume each g_n is entire and $g_n \rightarrow g$ uniformly on compact sets in $\mathbb{C}$ (i.e. $g_n \rightarrow g$ in $H(\mathbb{C})$). If each g_n has only real zeros, show that g has only real zeroes.
7. Assume that f and g are analytic and nonvanishing on $\{z \in \mathbb{C} : |z| < 2\}$ and that $|f(z)| = |g(z)|$ when $|z| = 1$. Show that there is a constant $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $f(z) = \lambda g(z)$ for all $z \in B_2(0)$.
8. Assume that f is entire, and for some integer $n > 0$,

$$\lim_{z \rightarrow \infty} \frac{f(z)}{z^n} = c,$$

for some finite, nonzero complex number c . What can you say about f ?